Recall:

1. $x(t) \overset{F}{\longleftrightarrow} X(\omega)$

$$x(t)\cdot e^{j\omega_0 t} \overset{F}{\longleftrightarrow} X(\omega-\omega_0)$$

2. $\alpha x(t) + \beta y(t) \overset{F}{\longleftrightarrow} \alpha X(\omega) + \beta Y(\omega)$.

3. $x(t)\cdot \cos\omega_0 t = \frac{1}{2} x(t)\cdot e^{j\omega_0 t} + \frac{1}{2} x(t)\cdot e^{-j\omega_0 t}$

$$\Big\updownarrow F$$

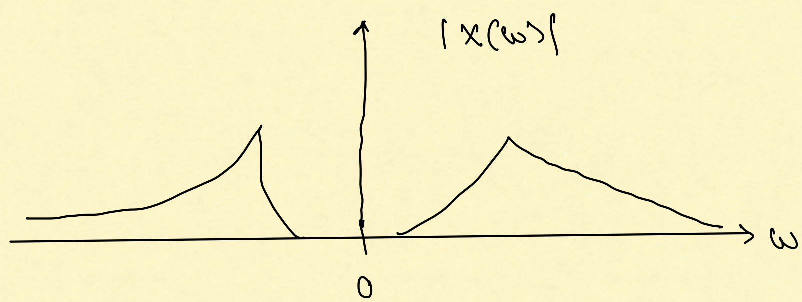
$$\frac{1}{2} X(\omega-\omega_0) + \frac{1}{2} X(\omega+\omega_0).$$

4. If $x(t)$ is real valued:

$$X(-\omega) = X^*(\omega) \quad \forall \omega.$$

$$|X(-\omega)| = |X^*(\omega)| = |X(\omega)|$$

$\therefore$ $|X(\omega)|$ is an even function.



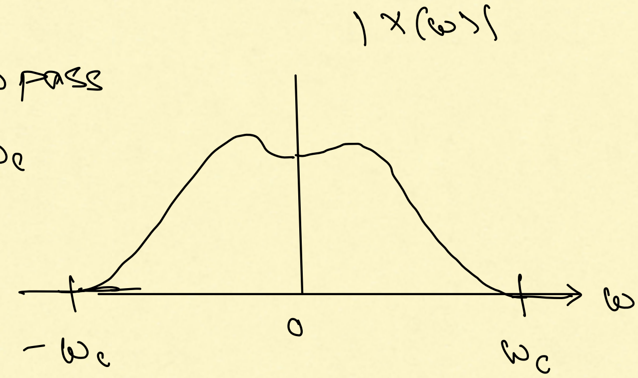
I. Bandwidth of Real-Valued Signals.

We will assume that $x(t)$ is real-valued.

$$\Rightarrow X(-\omega) = X^*(\omega).$$

Defn: A low pass signal

We say that $x(t)$ is a low pass signal if there exists an ω_c such that



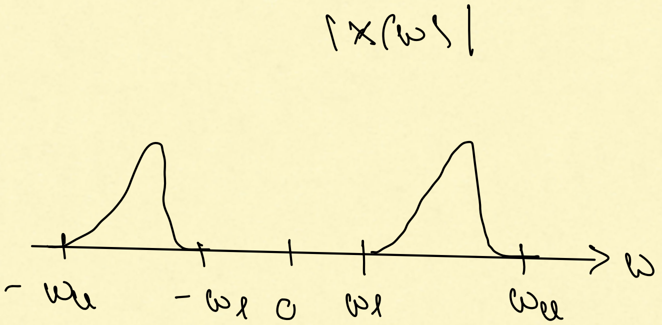
$X(\omega) = 0$ for

$\omega \geq \omega_c$ and $\omega \leq -\omega_c$.

The smallest such ω_c is called the bandwidth of the signal $x(t)$.

Defn: We say that a signal is a band pass signal if there exist $0 < \omega_l < \omega_u$ such that

For all $\omega \geq 0$ we have:



$X(\omega) = 0$ if

$0 \leq \omega \leq \omega_l$

and $\omega \geq \omega_u$.

The largest such ω_l is called lower cut-off frequency

The smallest " ω_u " " upper cut-off frequency

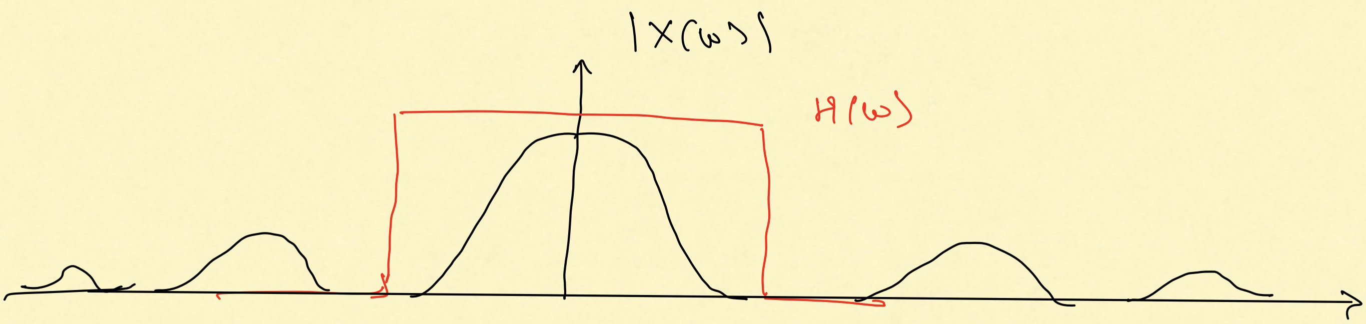
Band width of $x(t)$ is $(\omega_u - \omega_l)$ rad/s.

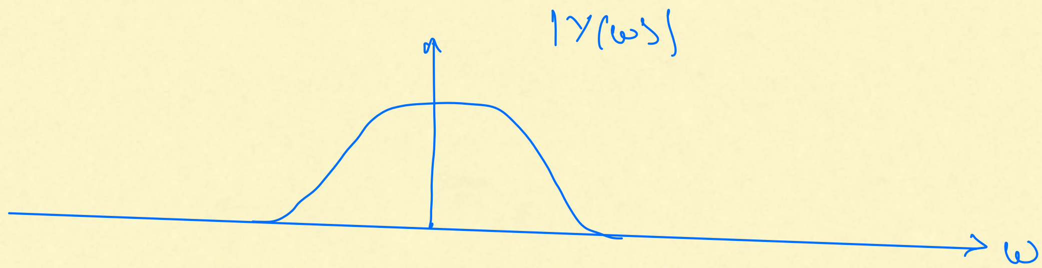
II. Filters.

$$x \longrightarrow \boxed{h} \longrightarrow y$$

where x and h are real-valued.

$$Y(\omega) = X(\omega) \cdot H(\omega).$$





Defn: An ideal lowpass filter (LPF) is an LTI system with frequency response or transfer function $H(\omega)$:

$$H(\omega) = \begin{cases} 1 & \text{if } -\omega_c < \omega < \omega_c \\ 0 & \text{if } |\omega| \geq \omega_c. \end{cases}$$

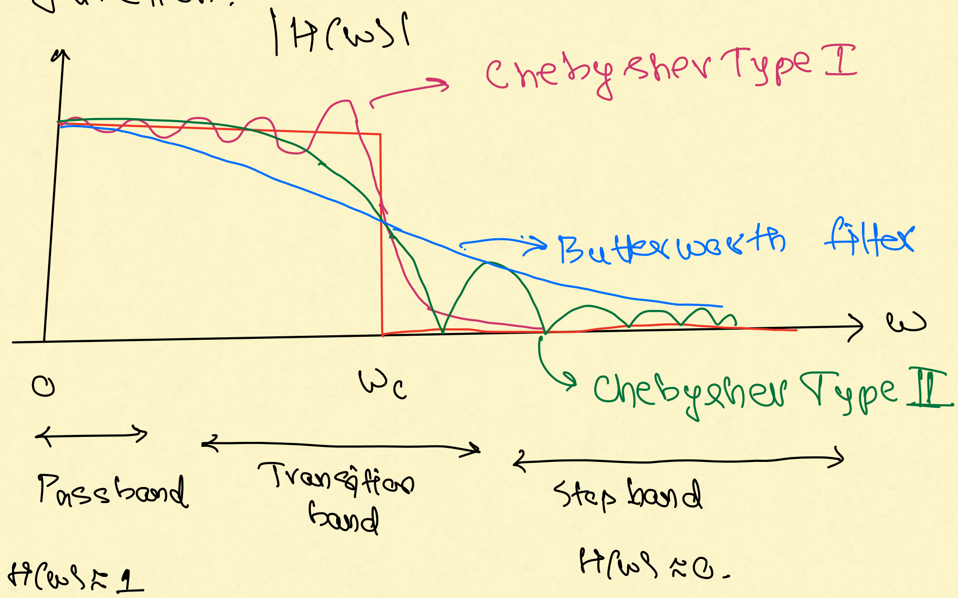
Defn: An ideal band pass filter (BPF) is an LTI system with transfer function:

$$H(\omega) = \begin{cases} 1 & \text{if } \omega_l < \omega < \omega_u \text{ or } -\omega_u < \omega < -\omega_l \\ 0 & \text{otherwise.} \end{cases}$$

Defn: Similar definition for ideal high pass filter (HPF).

Example: Practical filters do not have this ideal transfer function.

Consider LPF



Practical Design Considerations:

1. Transition band must be as narrow as possible.

2. Passband gain must be as close to 1 as possible. Ripples, if present, must be minimal.

3. The stop band gain must be as small as possible.

Apart from these, there are other considerations as well, such as rise time (in step response), ringing and group delay.